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LIST OF SYMBOLS, ABBREVIATIONS AND NOMENCLATURE

Abbreviation	Nomenclature
AICTE	All India Council for Technical Education
API	Application Programming Interface
CSBS	Computer Science and Business Systems
DFD	Data Flow Diagram
DTE	Directorate of Technical Education
ERD	Entity Relationship Diagram
FAO	Food and Agriculture Organization
FCM	Firebase Cloud Messaging
FK	Foreign Key
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
ICT	Information and Communication Technology
JSON	JavaScript Object Notation
JWT	JSON Web Token
PSO	Program Specific Outcomes
REST	Representational State Transfer
SQL	Structured Query Language
SSL	Secure Sockets Layer
UPI	Unified Payments Interface

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INTRODUCTION

CHAPTER 2

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ABSTRACT

Agriculture forms the backbone of the Indian economy, yet the sector remains plagued by systemic inefficiencies that severely marginalize farmers. Cultivators consistently grapple with volatile market prices, an over-reliance on exploitative intermediaries, and a profound lack of direct access to reliable buyers. These fractured supply chains not only diminish farmers' profit margins but also create a high-risk environment that stifles sustainable agricultural growth. There is an urgent need for a transparent, decentralized ecosystem capable of bridging the gap between production and procurement to ensure financial stability for the farming community.

To address this critical market gap, we are developing KrishiSetu, a digital contract-farming platform designed to revolutionize agricultural procurement. Operating initially as a robust Minimum Viable Product (MVP), KrishiSetu functions as a transparent marketplace where verified institutional buyers such as retailers, distributors, and agri-businesses can post detailed farming contracts specifying crop requirements, fixed price agreements, and delivery timelines. Farmers can intuitively explore, filter, and accept these pre-sowing contracts based on their geographical location and production capacity. By facilitating direct, legally sound agreements before cultivation even begins, KrishiSetu guarantees stable market access, eliminates predatory middlemen, and builds a foundation of predictable income.

Built for scale and inclusivity, KrishiSetu is engineered specifically to bridge the digital divide in rural areas. The MVP is accessible via a comprehensive web dashboard and a cross-platform Android application developed using React Native. Powered by a highly optimized Node.js backend, the architecture ensures seamless performance and minimal data consumption, making the platform fully operational even on low-end smartphones and basic PCs. Furthermore, the system is designed with a forward-looking, modular framework. Future iterations slated for open-source development will integrate advanced, data-driven modules including AI-powered regional price prediction and computer-vision-based crop disease detection evolving KrishiSetu from a transactional marketplace into a comprehensive, technology-empowered agricultural ecosystem.